Partial Fractions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Distinct linear factors**

Express $(5x + 7)/((x + 1)(x + 2))$ in partial fractions.

Write $(5x + 7)/((x + 1)(x + 2)) = A/(x + 1) + B/(x + 2)$. Multiply through: $5x + 7 = A(x + 2) + B(x + 1)$. At $x = -1$: $2 = A(1)$, so $A = 2$. At $x = -2$: $-3 = B(-1)$, so $B = 3$. Result: $2/(x + 1) + 3/(x + 2)$.

2 **[Foundational]** • **Target: Distinct linear factors**

Express $(3x - 1)/(x(x - 3))$ in partial fractions.

Write $(3x - 1)/(x(x - 3)) = A/x + B/(x - 3)$. Then $3x - 1 = A(x - 3) + Bx$. At $x = 0$: $-1 = -3A$, so $A = 1/3$. At $x = 3$: $8 = 3B$, so $B = 8/3$. Result: $1/(3x) + 8/(3(x - 3))$.

3 **[Foundational]** • **Target: Repeated linear factors**

Express $(4x + 1)/(x - 1)^2$ in partial fractions of the form $A/(x - 1) + B/(x - 1)^2$.

$4x + 1 = A(x - 1) + B$. At $x = 1$: $5 = B$. Comparing coefficients of x: $A = 4$. Result: $4/(x - 1) + 5/(x - 1)^2$.

4 **[Foundational]** • **Target: Repeated linear factors**

Express $(2x^2 - 3x + 1)/(x(x - 1)^2)$ in partial fractions of the form $A/x + B/(x - 1) + C/(x - 1)^2$.

$2x^2 - 3x + 1 = A(x - 1)^2 + Bx(x - 1) + Cx$. At $x = 0$: $1 = A(1)$, so $A = 1$. At $x = 1$: $0 = C(1)$, so $C = 0$. Comparing x^2 coefficients: $2 = A + B$, so $B = 2 - 1 = 1$. Result: $1/x + 1/(x - 1)$.

5 **[Foundational]** • **Target: Irreducible quadratic factors**

Express $(3x^2 + 2)/((x - 1)(x^2 + 1))$ in partial fractions of the form $A/(x - 1) + (Bx + C)/(x^2 + 1)$.

$3x^2 + 2 = A(x^2 + 1) + (Bx + C)(x - 1)$. At $x = 1$: $5 = 2A$, so $A = 5/2$. Comparing x^2 coefficients: $3 = A + B$, so $B = 3 - 5/2 = 1/2$. Comparing constants: $2 = A - C$, so $C = A - 2 = 1/2$. Result: $5/(2(x - 1)) + (x + 1)/(2(x^2 + 1))$.

6 **[Foundational]** • **Target: Applying partial fractions to integration**

Use partial fractions to evaluate $\int (5x + 7)/((x + 1)(x + 2))\,dx$, using the decomposition found earlier in this worksheet.

From earlier: $(5x + 7)/((x + 1)(x + 2)) = 2/(x + 1) + 3/(x + 2)$. So $\int (5x + 7)/((x + 1)(x + 2))\,dx = 2\ln|x + 1| + 3\ln|x + 2| + C$.

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7 [Foundational] • Target: Applying partial fractions to a binomial series expansion

Express $1/((1-x)(1-2x))$ in partial fractions, then use the result to write the first three terms of its series expansion in ascending powers of x.

$1/((1-x)(1-2x)) = A/(1-x) + B/(1-2x)$. Then $1 = A(1-2x) + B(1-x)$. At $x = 1$: $1 = A(-1)$, so $A = -1$. At $x = 1/2$: $1 = B(1/2)$, so $B = 2$. Result: $-1/(1-x) + 2/(1-2x)$. Expanding each geometric series: $-1(1 + x + x^2 + \cdots) + 2(1 + 2x + 4x^2 + \cdots) = (-1+2) + (-1+4)x + (-1+8)x^2 + \cdots = 1 + 3x + 7x^2 + \cdots$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Combining terms before decomposing

Express $(7-x)/((x+1)(x-2))$ in partial fractions.

Write $(7-x)/((x+1)(x-2)) = A/(x+1) + B/(x-2)$. Then $7 - x = A(x-2) + B(x+1)$. At $x = -1$: $8 = A(-3)$, so $A = -8/3$. At $x = 2$: $5 = B(3)$, so $B = 5/3$. Result: $-8/(3(x+1)) + 5/(3(x-2))$.

9 [Intermediate] • Target: Combining terms before decomposing

Express $(x^2 + x + 1)/(x(x+1))$ in partial fractions. (Hint: the degree of the numerator equals the degree of the denominator, so first perform polynomial division.)

Since numerator and denominator are both degree 2, divide first: $(x^2 + x + 1)/(x^2 + x) = 1 + 1/(x^2 + x) = 1 + 1/(x(x+1))$. Decompose $1/(x(x+1)) = A/x + B/(x+1)$: $1 = A(x+1) + Bx$. At $x = 0$: $A = 1$. At $x = -1$: $B = -1$. Result: $1 + 1/x - 1/(x+1)$.

10 [Intermediate] • Target: Applying partial fractions to a rate problem

A population model gives the rate of change $dP/dt = 1/(P(10-P))$. Express $1/(P(10-P))$ in partial fractions of the form $A/P + B/(10-P)$.

$1 = A(10-P) + BP$. At $P = 0$: $1 = 10A$, so $A = 1/10$. At $P = 10$: $1 = 10B$, so $B = 1/10$. Result: $1/(10P) + 1/(10(10-P))$ — a decomposition commonly used to solve logistic-growth differential equations by separation of variables.

11 [Intermediate] • Target: Partial fractions with three distinct linear factors

Express $(6x^2 - 2x - 1)/((x-1)(x+1)(x-2))$ in partial fractions.

Write $(6x^2 - 2x - 1)/((x-1)(x+1)(x-2)) = A/(x-1) + B/(x+1) + C/(x-2)$. Then $6x^2 - 2x - 1 = A(x+1)(x-2) + B(x-1)(x-2) + C(x-1)(x+1)$. At $x = 1$: $3 = A(2)(-1) = -2A$, so $A = -3/2$. At $x = -1$: $7 = B(-2)(-3) = 6B$, so $B = 7/6$. At $x = 2$: $19 = C(1)(3) = 3C$, so $C = 19/3$.

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12 **[Intermediate]** • **Target: Partial fractions with an improper fraction and repeated factor**

Express $x^3/(x-1)^2$ in partial fractions, first performing polynomial division since the numerator has higher degree than the denominator.

Divide x^3 by $(x-1)^2 = x^2 - 2x + 1$: $x^3 \div (x^2 - 2x + 1) = x + 2$ remainder $3x - 2$. So $x^3/(x-1)^2 = x + 2 + (3x-2)/(x-1)^2$. Decompose the remainder: $(3x-2)/(x-1)^2 = A/(x-1) + B/(x-1)^2$, giving $3x - 2 = A(x-1) + B$. At $x = 1$: $1 = B$. Comparing x coefficients: $A = 3$. Result: $x + 2 + 3/(x-1) + 1/(x-1)^2$.

13 **[Intermediate]** • **Target: Using partial fractions to sum a series**

Express $1/(n(n+1))$ in partial fractions, and use the result to find $1/(1 \cdot 2) + 1/(2 \cdot 3) + 1/(3 \cdot 4) + 1/(4 \cdot 5)$ by telescoping.

$1/(n(n+1)) = 1/n - 1/(n+1)$ (standard decomposition with $A = 1, B = -1$). Summing from $n = 1$ to 4: $(1 - 1/2) + (1/2 - 1/3) + (1/3 - 1/4) + (1/4 - 1/5) = 1 - 1/5 = 4/5$, since all the middle terms cancel (telescope).

14 **[Intermediate]** • **Target: Decomposing a fraction with a repeated quadratic-adjacent structure**

Express $(x+5)/((x-1)(x+1)^2)$ in partial fractions of the form $A/(x-1) + B/(x+1) + C/(x+1)^2$.

$x + 5 = A(x+1)^2 + B(x-1)(x+1) + C(x-1)$. At $x = 1$: $6 = A(4)$, so $A = 3/2$. At $x = -1$: $4 = C(-2)$, so $C = -2$. Comparing x^2 coefficients: $0 = A + B$, so $B = -3/2$. Result: $3/(2(x-1)) - 3/(2(x+1)) - 2/(x+1)^2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Decomposing a fraction with a repeated quadratic-adjacent structure** • **Cross-Lesson: 1.01 Proof by Mathematical Induction**

Express $(2x^2 + x - 1)/(x^2(x+1))$ in partial fractions of the form $A/x + B/(x^2) + C/(x+1)$.

$2x^2 + x - 1 = Ax(x+1) + B(x+1) + Cx^2$. At $x = 0$: $-1 = B(1)$, so $B = -1$. At $x = -1$: $0 = C(1)$, so $C = 0$. Comparing x^2 coefficients: $2 = A + C = A$, so $A = 2$. Result: $2/x - 1/(x^2)$.

16 **[Challenging]** • **Target: Decomposing a fraction with a repeated quadratic-adjacent structure** • **Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra**

Use partial fractions to find the Maclaurin (power series) expansion of $1/((1-x)(2-x))$ up to the term in x^2, first decomposing into the form $A/(1-x) + B/(2-x)$.

$1 = A(2-x) + B(1-x)$. At $x = 1$: $1 = A(1)$, so $A = 1$. At $x = 2$: $1 = B(-1)$, so $B = -1$. Result: $1/(1-x) - 1/(2-x) = 1/(1-x) - 1/2 \cdot 1/(1 - x/2)$. Expanding each as a geometric series:

$(1 + x + x^2 + \cdots) - 1/2(1 + x/2 + x^2/4 + \cdots) = (1 - 1/2) + (1 - 1/4)x + (1 - 1/8)x^2 + \cdots = 1/2 + 3/4x + 7/8x^2 + \cdots$.

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17 **[Challenging]** • **Target: Decomposing a fraction with a repeated quadratic-adjacent structure** • **Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity**

Express $(4x^2 - 5x - 15)/((x+1)(x-2)(x+3))$ in partial fractions, using the cover-up method (evaluating the numerator at each root while covering the matching factor in the denominator).

Write $(4x^2 - 5x - 15)/((x+1)(x-2)(x+3)) = A/(x+1) + B/(x-2) + C/(x+3)$. Using the cover-up method:

$A = (4(-1)^2 - 5(-1) - 15)/((-1-2)(-1+3)) = (4+5-15)/((-3)(2)) = -6/(-6) = 1$. $B = (4(2)^2 - 5(2) - 15)/((2+1)(2+3)) = (16-10-15)/((3)(5)) = -9/15 = -3/5$.

$C = (4(-3)^2 - 5(-3) - 15)/((-3+1)(-3-2)) = (36+15-15)/((-2)(-5)) = 36/10 = 18/5$. Result: $1/(x+1) - 3/(5(x-2)) + 18/(5(x+3))$.

18 **[Challenging]** • **Target: Decomposing a fraction with a repeated quadratic-adjacent structure** • **Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)**

A chemical reaction's rate is modeled by $dC/dt = 1/((C+2)(C-1))$, where C is a concentration. Express $1/((C+2)(C-1))$ in partial fractions of the form $A/(C+2) + B/(C-1)$, and then use the result to write the general antiderivative $\int 1/((C+2)(C-1))\,dC$ in terms of natural logarithms.

$1 = A(C-1) + B(C+2)$. At $C = 1$: $1 = B(3)$, so $B = 1/3$. At $C = -2$: $1 = A(-3)$, so $A = -1/3$. Result: $-1/3/(C+2) + 1/3/(C-1)$. Integrating: $\int 1/((C+2)(C-1))\,dC = -1/3\ln|C+2| + 1/3\ln|C-1| + K = 1/3\ln|(C-1)/(C+2)| + K$.

19 **[Challenging]** • **Target: Decomposing a fraction with a repeated quadratic-adjacent structure** • **Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem**

Use the decomposition $1/(3x) + 8/(3(x-3))$ found earlier in this worksheet for $(3x-1)/(x(x-3))$ to evaluate $\int_4^5 (3x-1)/(x(x-3))\,dx$, giving your answer as a single logarithm expression.

$\int (3x-1)/(x(x-3))\,dx = 1/3\ln|x| + 8/3\ln|x-3| + C$. Evaluating from 4 to 5:

$[1/3\ln 5 + 8/3\ln 2] - [1/3\ln 4 + 8/3\ln 1] = 1/3\ln 5 + 8/3\ln 2 - 1/3\ln 4 = 1/3\ln(5/4) + 8/3\ln 2$.

20 **[Challenging]** • **Target: Decomposing a fraction with a repeated quadratic-adjacent structure** • **Cross-Lesson: 3.01 3D Vectors & the Cross Product**

This question has three parts. (a) Factor $x^2 - 4$ and hence express $(x+8)/(x^2-4)$ in partial fractions. (b) Use your result from part (a) to write down $\int (x+8)/(x^2-4)\,dx$. (c) Hence evaluate $\int_3^5 (x+8)/(x^2-4)\,dx$, giving your answer as a single expression involving natural logarithms.

(a) Factor $x^2 - 4 = (x-2)(x+2)$, so $(x+8)/((x-2)(x+2)) = A/(x-2) + B/(x+2)$. Then $x + 8 = A(x+2) + B(x-2)$. At $x = 2$: $10 = 4A$, so $A = 5/2$. At $x = -2$: $6 = -4B$, so $B = -3/2$. Result: $5/2/(x-2) - 3/2/(x+2)$. (b) $\int (x+8)/(x^2-4)\,dx = 5/2\ln|x-2| - 3/2\ln|x+2| + C$. (c) Evaluating from 3 to 5: $[5/2\ln 3 - 3/2\ln 7] - [5/2\ln 1 - 3/2\ln 5] = 5/2\ln 3 - 3/2\ln 7 + 3/2\ln 5$ (since $\ln 1 = 0$).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Identifying the correct form of a decomposition

Which of the following is the correct form for the partial fraction decomposition of $(3x+1)/((x-2)^2(x+1))$?

a. **A/(x-2)+B/(x-2)^2+C/(x+1)** ✓

b. A/(x-2)+B/(x+1)

c. A/((x-2)^2)+B/(x+1)

d. (Ax+B)/(x-2)^2+C/(x+1)

A repeated linear factor $(x-2)^2$ requires two separate terms, $A/(x-2)$ and $B/(x-2)^2$, while the distinct linear factor $(x+1)$ contributes one term, $C/(x+1)$. The second option omits the repeated-factor structure, the third omits the lower-power term, and the fourth incorrectly uses a linear numerator over a repeated linear factor (that form is only needed for irreducible quadratic factors).

22 [Foundational] • Target: Distinct Linear Factors

Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions $\frac{A}{x-1}+\frac{B}{x+2}$.

a. $A=\frac{4}{3}, B=\frac{11}{3}$ ✓

b. $A=4, B=11$

c. $A=1, B=4$

d. $A=2, B=3$

Cover-up rule: $A=\frac{5x-1}{x+2}\Big|_{x=1}=\frac{4}{3}$. $B=\frac{5x-1}{x-1}\Big|_{x=-2}=\frac{-11}{-3}=\frac{11}{3}$.

23 [Foundational] • Target: Distinct Linear Factors

Find the partial fraction decomposition of $\frac{1}{x^2-9}$.

a. $\frac{1/6}{x-3}-\frac{1/6}{x+3}$ ✓

b. $\frac{1/3}{x-3}+\frac{1/3}{x+3}$

c. $\frac{1}{x-3}-\frac{1}{x+3}$

d. $\frac{1/6}{x-3}+\frac{1/6}{x+3}$

$\frac{1}{(x-3)(x+3)}=\frac{A}{x-3}+\frac{B}{x+3}$. $A=\frac{1}{x+3}\Big|_{x=3}=\frac{1}{6}$. $B=\frac{1}{x-3}\Big|_{x=-3}=-\frac{1}{6}$.

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24 [Foundational] • Target: Distinct Linear Factors

Evaluate $\int \frac{3}{(x+1)(x+4)}dx$.

a. $\ln\left|\frac{x+1}{x+4}\right| + C$ ✓

b. $3\ln|x+1| + 3\ln|x+4| + C$

c. $\frac{1}{3}\ln\left|\frac{x+1}{x+4}\right| + C$

d. $\ln|(x+1)(x+4)| + C$

Decompose: $\frac{3}{(x+1)(x+4)} = \frac{1}{x+1} - \frac{1}{x+4}$. Integrating: $\ln|x+1| - \ln|x+4| + C = \ln|\frac{x+1}{x+4}| + C$.

25 [Foundational] • Target: Repeated Linear Factors

What is the correct template for decomposing $\frac{2x+7}{(x-2)^2(x+1)}$?

a. $\frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+1}$ ✓

b. $\frac{A}{x-2} + \frac{B}{x+1}$

c. $\frac{Ax+B}{(x-2)^2} + \frac{C}{x+1}$

d. $\frac{A}{(x-2)^2} + \frac{B}{x+1}$

A repeated linear factor $(x-c)^2$ requires terms for each ascending power: $\frac{A}{x-c} + \frac{B}{(x-c)^2}$, plus terms for any distinct factors.

26 [Foundational] • Target: Repeated Linear Factors

In $\frac{3x^2+1}{(x-1)^2} = A + \frac{B}{x-1} + \frac{C}{(x-1)^2}$, find C.

a. 4 ✓

b. 3

c. 1

d. 0

Multiply through by $(x-1)^2$: $3x^2+1 = A(x-1)^2 + B(x-1) + C$. Setting $x = 1$ yields $3(1)^2 + 1 = C \Rightarrow C = 4$.

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27 [Foundational] • Target: Repeated Linear Factors

Decompose $\frac{x}{(x+1)^2}$.

a. $\frac{1}{x+1} - \frac{1}{(x+1)^2}$ ✓

b. $\frac{1}{x+1} + \frac{1}{(x+1)^2}$

c. $\frac{x}{x+1} - \frac{1}{(x+1)^2}$

d. $\frac{2}{x+1} - \frac{1}{(x+1)^2}$

Rewrite numerator: $x = (x+1) - 1$. Then $\frac{(x+1)-1}{(x+1)^2} = \frac{1}{x+1} - \frac{1}{(x+1)^2}$.

28 [Foundational] • Target: Irreducible Quadratic Factors

What partial fraction form should be set up for $\frac{4x^2+2x+5}{(x-3)(x^2+4)}$?

a. $\frac{A}{x-3} + \frac{Bx+C}{x^2+4}$ ✓

b. $\frac{A}{x-3} + \frac{B}{x^2+4}$

c. $\frac{A}{x-3} + \frac{B}{x+2} + \frac{C}{x-2}$

d. $\frac{Ax+B}{x-3} + \frac{Cx+D}{x^2+4}$

An irreducible real quadratic factor $(x^2 + a^2)$ requires a linear numerator $(Bx + C)$ over the quadratic denominator.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Irreducible Quadratic Factors

In the decomposition $\frac{x^2+1}{(x-2)(x^2+1)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1}$, find A.

a. $\frac{5}{5} = 1$ ✓

b. 0

c. 2

d. -1

Notice $\frac{x^2+1}{(x-2)(x^2+1)} = \frac{1}{x-2}$. Thus $A = 1$ and $B = C = 0$.

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30 [Intermediate] • Target: Irreducible Quadratic Factors

Evaluate $\int \frac{2x+3}{x^2+9}dx$.

a. $\ln(x^2+9)+\arctan\left(\frac{x}{3}\right)+C$ ✓

b. $\ln(x^2+9)+3\arctan(x)+C$

c. $\frac{1}{2}\ln(x^2+9)+\frac{1}{3}\arctan(x/3)+C$

d. $2\ln(x^2+9)+C$

Split into two integrals: $\int \frac{2x}{x^2+9}dx+\int \frac{3}{x^2+9}dx=\ln(x^2+9)+3\cdot\frac{1}{3}\arctan\left(\frac{x}{3}\right)+C=\ln(x^2+9)+\arctan\left(\frac{x}{3}\right)+C$.

31 [Intermediate] • Target: Improper Rational Fractions (Long Division First)

Before performing partial fractions on $\frac{x^3+2x^2+1}{x^2-1}$, what preliminary algebraic step is mandatory?

a. **Polynomial long division to separate into polynomial quotient plus proper remainder** ✓

b. Factoring the numerator completely

c. Applying the cover-up method directly

d. Multiplying numerator and denominator by x

Because the degree of the numerator (3) is greater than or equal to the degree of the denominator (2), long division is required first.

32 [Intermediate] • Target: Improper Rational Fractions (Long Division First)

Perform polynomial long division on $\frac{x^3+2x^2+1}{x^2-1}$. What is the quotient?

a. $x+2$ ✓

b. $x-2$

c. x^2+2

d. x

$x(x^2-1)=x^3-x$. Subtract: $2x^2+x+1$. Next term: $2(x^2-1)=2x^2-2$. Subtract: $x+3$. The quotient is $x+2$ with remainder $x+3$.

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33 [Intermediate] • Target: Improper Rational Fractions (Long Division First)

Decompose the remainder term $\frac{x+3}{x^2-1} = \frac{x+3}{(x-1)(x+1)}$.

a. $\frac{2}{x-1} - \frac{1}{x+1}$ ✓

b. $\frac{1}{x-1} + \frac{2}{x+1}$

c. $\frac{3}{x-1} - \frac{2}{x+1}$

d. $\frac{1}{x-1} - \frac{1}{x+1}$

$A = \frac{x+3}{x+1}\Big|_{x=1} = \frac{4}{2} = 2$. $B = \frac{x+3}{x-1}\Big|_{x=-1} = \frac{2}{-2} = -1$. Thus $\frac{2}{x-1} - \frac{1}{x+1}$.

34 [Intermediate] • Target: Solving a separable differential equation using partial fractions

Consider the quadratic function $f(x) = x^2 - 4x + 3$ and the linear function $g(x) = x - 1$. The points of intersection are given by their solutions. What is the y-coordinate of the point of intersection where x = 1?

a. **0** ✓

b. 1

c. -1

d. 2

To find the points of intersection, we set $f(x) = g(x)$: $x^2 - 4x + 3 = x - 1$. $x^2 - 5x + 4 = 0$. $(x-1)(x-4) = 0$. The x-coordinates of the intersection points are $x = 1$ and $x = 4$. We are asked for the y-coordinate when $x = 1$. We can use either function to find this y-coordinate. Using $g(x) = x - 1$, when $x = 1$, $g(1) = 1 - 1 = 0$. Therefore, the y-coordinate is 0.

35 [Intermediate] • Target: Explaining when polynomial division is required

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points where $f(x) = 0$. Determine the sum of the x-coordinates of these intersection points.

a. 1

b. 3

c. **4** ✓

d. -3

The x-intercepts of the graph of $y = f(x)$ are the roots of the equation $f(x) = 0$. For a quadratic equation of the form $ax^2 + bx + c = 0$, the sum of the roots is given by $-\frac{b}{a}$. In this case, $f(x) = x^2 - 4x + 3$, so $a = 1$, $b = -4$, and $c = 3$. The sum of the x-coordinates of the intersection points is $-\frac{-4}{1} = 4$.

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36 [Intermediate] • Target: Repeated irreducible quadratic factor

The graph of the quadratic function $f(x) = x^2 - 5x + 6$ is shown below. The graph intersects the x-axis at points A and B. Find the sum of the x-coordinates of points A and B.

a. 5 ✓

b. -5

c. 6

d. -6

The x-coordinates of points A and B are the roots of the quadratic equation $f(x) = 0$, which is $x^2 - 5x + 6 = 0$. For a quadratic equation of the form $ax^2 + bx + c = 0$, the sum of the roots is given by $-\frac{b}{a}$. In this equation, $a = 1$ and $b = -5$. Therefore, the sum of the roots is $-\frac{-5}{1} = 5$.

37 [Intermediate] • Target: Verifying a decomposition by substitution

The diagram shows the graph of the function $y = f(x)$. The equation of the function is given by $f(x) = ax^2 + bx + c$. Given that the vertex of the parabola is at $(-1, 8)$ and the graph passes through the point $(1, 0)$, find the values of a and b.

a. a = -2, b = -4 ✓

b. a = -2, b = 4

c. a = 2, b = 4

d. a = 2, b = -4

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ is at $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$. We are given that the vertex is at $(-1, 8)$. Therefore, $-\frac{b}{2a} = -1$, which implies $b = 2a$. We are also given that the vertex is at $y = 8$, so $f(-1) = 8$. Substituting $x = -1$ into the function: $a(-1)^2 + b(-1) + c = 8 \Rightarrow a - b + c = 8$. Since $b = 2a$, we have $a - 2a + c = 8 \Rightarrow -a + c = 8$. The graph also passes through the point $(1, 0)$. So, $f(1) = 0$. Subst.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Choosing efficient substitution values • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Consider the quadratic equation $x^2 - kx + 8 = 0$, where k is a real parameter. The graph of the corresponding quadratic function $y = x^2 - kx + 8$ is a parabola. Given that the parabola intersects the x-axis at exactly one point, find the value of k.

a. $-4\sqrt{2}$

b. $-2\sqrt{2}$

c. $2\sqrt{2}$

d. $4\sqrt{2}$ ✓

The quadratic equation $ax^2 + bx + c = 0$ has exactly one real root (i.e., the parabola $y = ax^2 + bx + c$ intersects the x-axis at exactly one point) if and only if its discriminant, $\Delta = b^2 - 4ac$, is equal to zero. In the given equation, $x^2 - kx + 8 = 0$, we have $a = 1$, $b = -k$, and $c = 8$. Therefore, the discriminant is $\Delta = (-k)^2 - 4(1)(8) = k^2 - 32$. For the parabola to intersect the x-axis at exactly one point, we must have $\Delta = 0$. So, $k^2 - 32 = 0$. This gives $k^2 = 32$, which m.

39 [Challenging] • Target: Distinct Linear Factors • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

A random variable X follows a binomial distribution $B(n, p)$. The number of trials is $n = 10$ and the probability of success in a single trial is p. If the mean of X is 4, find the variance of X.

a. **2.4** ✓

b. 3.6

c. 6.4

d. 7.2

For a binomial distribution $X \sim B(n, p)$, the mean is given by $E(X) = np$ and the variance is given by $Var(X) = np(1 - p)$. We are given that $n = 10$ and $E(X) = 4$. Using the formula for the mean, we have $10p = 4$. Solving for p, we get $p = \frac{4}{10} = 0.4$. Now we can find the variance using the formula $Var(X) = np(1 - p)$. Substituting the values of n and p, we get $Var(X) = 10 \times 0.4 \times (1 - 0.4)$. $Var(X) = 10 \times 0.4 \times 0.6$. $Var(X) = 4 \times 0.6$. $Var(X) = 2.4$.

40 [Challenging] • Target: Repeated Linear Factors • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Find d in the equation $ax + by + cz = d$ of the plane through $A(-1, 3, 4)$, $B(-3, 1, 1)$, and $C(4, 2, -4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. -59

b. -46

c. -75

d. **-58** ✓

Working through this vectors step: $\vec{AB} = (-2; -2; -3)$, $\vec{AC} = (5; -1; -8)$, $\mathbf{n} = (13; -31; 12)$, $d = \mathbf{n} \cdot A = -58$

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41 [Challenging] • Target: Irreducible Quadratic Factors • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Find d in the equation $ax + by + cz = d$ of the plane through $A(2, -1, 4)$, $B(2, -3, -3)$, and $C(-4, 4, 2)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 0

b. -3

c. -6

d. **-12** ✓

Working through this vectors step: $\vec{AB} = (0; -2; -7)$, $\vec{AC} = (-6; 5; -2)$, $\mathbf{n} = (39; 42; -12)$, $d = \mathbf{n} \cdot A = -12$

42 [Challenging] • Target: Improper Rational Fractions (Long Division First) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Find d in the equation $ax + by + cz = d$ of the plane through $A(2, 0, 3)$, $B(-4, 0, 1)$, and $C(0, -2, -3)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 16

b. 12

c. 24

d. **28** ✓

Working through this vectors step: $\vec{AB} = (-6; 0; -2)$, $\vec{AC} = (-2; -2; -6)$, $\mathbf{n} = (-4; -32; 12)$, $d = \mathbf{n} \cdot A = 28$

43 [Challenging] • Target: Distinct linear factors • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Find d in the equation $ax + by + cz = d$ of the plane through $A(3, -1, -2)$, $B(-3, 2, 0)$, and $C(-1, 3, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 42

b. 32

c. **26** ✓

d. 22

Working through this vectors step: $\vec{AB} = (-6; 3; 2)$, $\vec{AC} = (-4; 4; 6)$, $\mathbf{n} = (10; 28; -12)$, $d = \mathbf{n} \cdot A = 26$

44 [Challenging] • Target: Repeated linear factors • Cross-Lesson: 4.02 The Poisson Distribution

Find d in the equation $ax + by + cz = d$ of the plane through $A(-4, -4, 1)$, $B(1, 0, -3)$, and $C(-4, 3, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. -100

b. -60

c. **-65** ✓

d. -45

Working through this vectors step: $\vec{AB} = (5; 4; -4)$, $\vec{AC} = (0; 7; 3)$, $\mathbf{n} = (40; -15; 35)$, $d = \mathbf{n} \cdot A = -65$

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45 [Challenging] • Target: Irreducible quadratic factors • Cross-Lesson: 5.01 Further Integration Techniques

Find d in the equation $ax + by + cz = d$ of the plane through $A(3, 1, -3)$, $B(0, -4, 3)$, and $C(3, 3, 0)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. **-54** ✓

b. -84

c. -63

d. -78

Working through this vectors step: $\vec{AB} = (-3; -5; 6),\ \vec{AC} = (0; 2; 3),\ \mathbf{n} = (-27; 9; -6),\ d = \mathbf{n} \cdot A = -54$

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